ASSIGNMENT 6

Question: - 1

Create a class Number Series that uses a loop to print numbers from 1 to 100.

Ans: - Code

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```python
class NumberSeries:
    def print_numbers(self):
        for x in range(1, 101):
            print(x)
series = NumberSeries()
series.print_numbers()
```

Output: -

1,2,3,4,5,6,7,8,9,10,.....,100

Question: - 2

Create a class EvenOddChecker that checks and prints whether numbers from 1 to 50 are even or odd using a loop.

Ans: - Code

```python
class EvenOddChecker:
    def check(self):
        for x in range(1, 51):
            if x % 2 == 0:
                print(x, "is Even Number.")
            else:
                print(x, "is Odd Number.")
obj = EvenOddChecker()
obj.check()
```

Output: -

```
1 is Odd Number.
2 is Even Number.
3 is Odd Number.
4 is Even Number.
5 is Odd Number.
6 is Even Number.
7 is Odd Number.
8 is Even Number.
9 is Odd Number.
10 is Even Number.
```

```
41 is Odd Number.
42 is Even Number.
43 is Odd Number.
44 is Even Number.
45 is Odd Number.
46 is Even Number.
47 is Odd Number.
48 is Even Number.
49 is Odd Number.
50 is Even Number.
PS D:\Data Science with AI & ML\PYTHON>
```

Question: - 3

Create a class MultiplicationTable with a function to print the multiplication table of a user-given number.

Ans: - Code

```
class MultiplicationTable:
    def table(self, x):
        for i in range(1, 11):
            print(x, "*", i, "=", x * i)
n = int(input("Enter a number: "))
obj = MultiplicationTable()
obj.table(n)
```

Output: -

```
Enter a number: 9
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
PS D:\Data Science with AI & ML\PYTHON>
```

Question: -4

Create a class FactorialCalculator with a function to calculate the factorial of a number using a loop.

Ans: - Code

```
class FactorialCalculator:
    def factorial(self, x):
        fact = 1
        for i in range(1, x + 1):
            fact = fact * i
        print("Factorial =", fact)
n = int(input("Enter a number:- "))
obj = FactorialCalculator()
obj.factorial(n)
```

Output: -

```
Enter a number:- 6
Factorial = 720
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```

Question: - 5

Create a class PrimeNumberChecker with a function to check whether a number is prime or not.

Ans: - Code

```
class PrimeNumber:
    def check(self, x):
        if x > 1:
            for i in range(2, x):
                if x % i == 0:
                    print("This is not Prime Number.",x)
                    break
            else:
                print("This is Prime Number.",x)
        else:
            print("This is not Prime Number.",x)
n = int(input("Enter a number: "))
obj = PrimeNumber()
obj.check(n)
```

Output: -

```
Enter a number: 5
This is Prime Number. 5
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```

```
Enter a number: 6
This is not Prime Number. 6
PS D:\Data Science with AI & ML\PYTHON>
```

Question: - 6

Create a class Student with attributes: a. name

b. roll number

c. course

Ans: - Code

```python
class Student:
        name = ""
        roll_number = 0
        course = ""
s1 = Student()
s1.name = "Aswhani Kumar"
s1.roll_number = 2
s1.course = "MCA"
print("Name:", s1.name)
print("Roll Number:", s1.roll_number)
print("Course:", s1.course)
```

Output: -

```
Name: Aswhani Kumar
Roll Number: 2
Course: MCA
PS D:\Data Science with AI & ML\PYTHON>
```

Question: - 7

Add a function to display student details.

Ans: - Code

```
class Student:
    name = ""
    roll_number = 0
    course = ""
    def get_details(self):
        self.name = input("Enter student name: ")
        self.roll_number = int(input("Enter roll number: "))
        self.course = input("Enter course: ")
    def display(self):
        print("\nStudent Details")
        print("Name:", self.name)
        print("Roll Number:", self.roll_number)
        print("Course:", self.course)
s = Student()
s.get_details()
s.display()
```

Output: -

```
Enter student name: Ashwani Kumar
Enter roll number: 2
Enter course: MCA

Student Details
Name: Ashwani Kumar
Roll Number: 2
Course: MCA
PS D:\Data Science with AI & ML\PYTHON\Assignment 6>
```

Question: - 8

Create a class Employee using a constructor (__init__) to store employee details and create a function to calculate yearly salary.

Ans: - Code

```python
class Employee:
    def __init__(self, name, emp_id, monthly_salary):
        self.name = name
        self.emp_id = emp_id
        self.monthly_salary = monthly_salary
    def yearly_salary(self):
        return self.monthly_salary * 12
    def display(self):
        print("\nEmployee Details")
        print("Name:", self.name)
        print("Employee ID:", self.emp_id)
        print("Monthly Salary:", self.monthly_salary)
        print("Yearly Salary:", self.yearly_salary())
emp = Employee("Ashwani Kumar", 2, 20000)
emp.display()
```

Output: -

```
Employee Details
Name: Ashwani Kumar
Employee ID: 2
Monthly Salary: 20000
Yearly Salary: 240000
PS D:\Data Science with AI & ML\PYTHON\Assignment 6>
```

Question: -9

Create a parent class Person and a child class Teacher using single inheritance. Display teacher details using inherited properties.

Ans: - Code

```python
class Person:
    def get_person(self):
        self.name = input("Enter name: ")
        self.age = int(input("Enter age: "))
class Teacher(Person):
    def get_teacher(self):
        self.subject = input("Enter subject: ")
    def display(self):
        print("\nTeacher Details")
        print("Name:", self.name)
        print("Age:", self.age)
        print("Subject:", self.subject)
t = Teacher()
t.get_person()
t.get_teacher()
t.display()
```

Output: -

```
Enter name: Ashwani Kumar
Enter age: 22
Enter subject: Python

Teacher Details
Name: Ashwani Kumar
Age: 22
Subject: Python
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```

Question: - 10

Create three classes Vehicle, Car, and SportsCar using multilevel inheritance and display information of a sports car.

Ans: - Code

```python
class Vehicle:
    def get_vehicle(self):
        self.brand = input("Enter vehicle brand: ")
class Car(Vehicle):
    def get_car(self):
        self.model = input("Enter car model: ")
class SportsCar(Car):
    def get_sportscar(self):
        self.speed = input("Enter top speed: ")
    def display(self):
        print("\nSports Car Details")
        print("Brand:", self.brand)
        print("Model:", self.model)
        print("Top Speed:", self.speed)
s = SportsCar()
s.get_vehicle()
s.get_car()
s.get_sportscar()
s.display()
```

Output: -

```
Enter vehicle brand: Mahindra
Enter car model: XUV7XO
Enter top speed: 240

Sports Car Details
Brand: Mahindra
Model: XUV7XO
Top Speed: 240
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```

Question: - 11

Create a class BankAccount with functions:

- deposit money
- withdraw money
- check balance

Ans: - Code

```python
class BankAccount:
        balance = 0
        def deposit(self):
            amount = float(input("Enter deposit amount:- "))
            self.balance = self.balance + amount
            print("Money Successfully Deposited.")
        def withdraw(self):
            amount = float(input("Enter withdraw amount:- "))
            if amount <= self.balance:
                self.balance = self.balance - amount
                print("Money Successfully Withdraw.")
            else:
                print("Insufficient Balance.")
        def check_balance(self):
            print("Current Balance:", self.balance)
b = BankAccount()
b.deposit()
b.withdraw()
b.check_balance()
```

Output: -

```
Enter deposit amount:- 500000
Money Successfully Deposited.
Enter withdraw amount:- 500000
Money Successfully Withdraw.
Current Balance: 0.0
PS D:\Data Science with AI & ML\PYTHON>
```